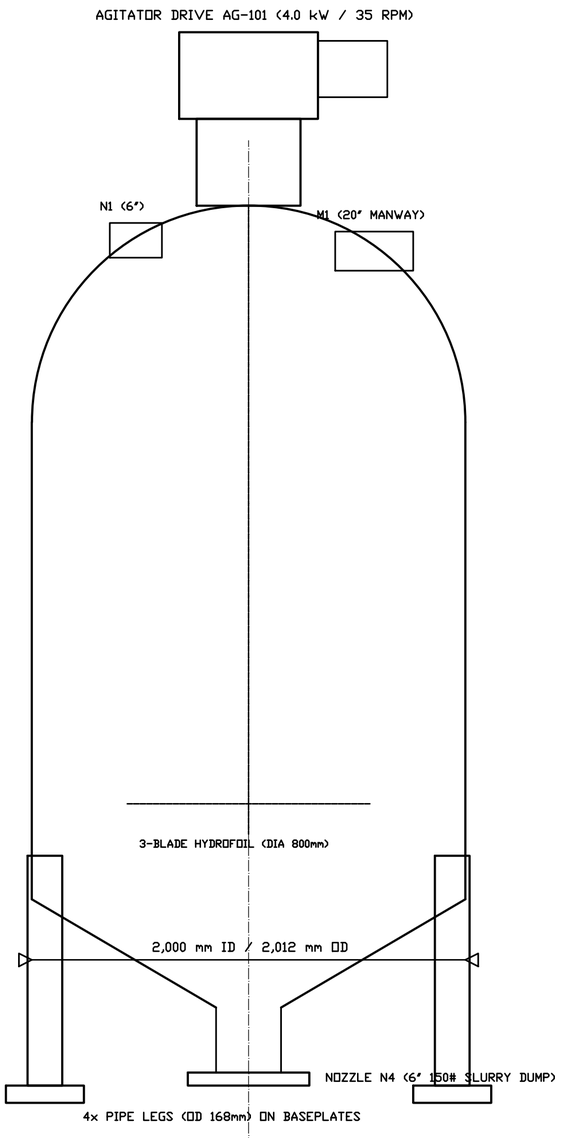
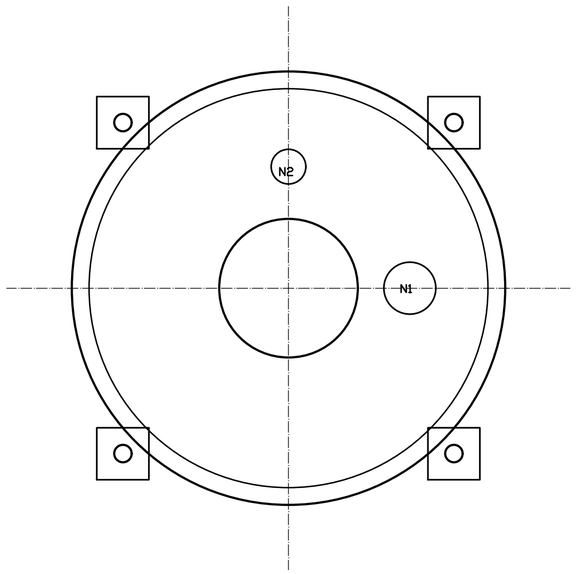


FRONT ELEVATION VIEW (SCALE 1:20)



TOP PLAN VIEW (SCALE 1:20)

MARK	SIZE	RATING	FACING	SERVICE / DESCRIPTION
N1	6" NB	150# ANSI	RF WNRF	Raw Dal Grain Infeed Chute
N2	2" NB	150# ANSI	RF WNRF	Process Soft Water Inflow
N3	1" NB	150# ANSI	RF WNRF	Chemical Reagent Dosing Inflow
N4	6" NB	150# ANSI	RF WNRF	Soaked Dal Slurry Dump to Screen
N5	2" NB	150# ANSI	RF WNRF	CIP Spray Ball Supply Header
N6	3" NB	150# ANSI	RF WNRF	Atmospheric Vent / Pressure Relief
M1	20" NB	150# ANSI	RF Davit	Inspection & Maintenance Manway

MECHANICAL DESIGN DATA & FABRICATION SPECIFICATION (ASME SEC VIII DIV 1)	
DESIGN PARAMETER	**ENGINEERING SPECIFICATION / VERIFIED VALUE**
Design Standard	ASME Boiler & Pressure Vessel Code Sec VIII Div 1 (App 14)
Working Capacity	7.50 m3 Net Liquid Volume | 8.45 m3 Gross Internal Volume
Operating Press & Temp	Atmospheric Hydrostatic (0.42 bar g head) | 30.0 deg C +/- 1 deg C
Design Press & Temp	Full Vacuum (-1.00 bar g) to +1.00 bar g | 65.0 deg C
Shell Metallurgy	AISI SS316L (6.0 mm nominal wall thk, 2B sanitary finish)
Head Metallurgy	AISI SS316L (8.0 mm thk, 2:1 Ellipsoidal seamless formed)
Conical Hopper	AISI SS316L (6.0 mm thk, 60 deg included angle smooth knuckle)
Internal Surface Finish	Ra <= 0.8 um (180 Grit electro-polished, welds ground flush)
External Finish	Descaled, acid passivated & Scotch-Brite uniform satin finish
Radiography & NDT	Spot Radiography per ASME UW-52 (Joint Efficiency E = 0.85)
Hydrostatic Test	Shop Hydro-tested at 1.30 x Design = 0.65 bar g held for 60 min
Agitator Geared Motor	SEW-Eurodrive Helical-bevel geared unit, 4.0 kW, 35 RPM output
Agitator Impeller	Dual 3-Blade high-efficiency hydrofoils (Dia 800 mm solid SS316L)
CIP Spray System	2x 360 deg Rotary spray balls (SS316L) covering 100% shell
Equipment Weights	Tare Empty: 1,850 kg | Operating Flooded: 9,350 kg
Foundation & Anchors	4x Pipe support legs, vertical load 23.4 kN/leg, 4x M20 SS304
Quality Certification	Manufacturer Data Report Form U-1 / EN 10204 3.1 Material Certs

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - SOAK VESSEL TK-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-007	REV: V8.0	SHEET: 7 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ